

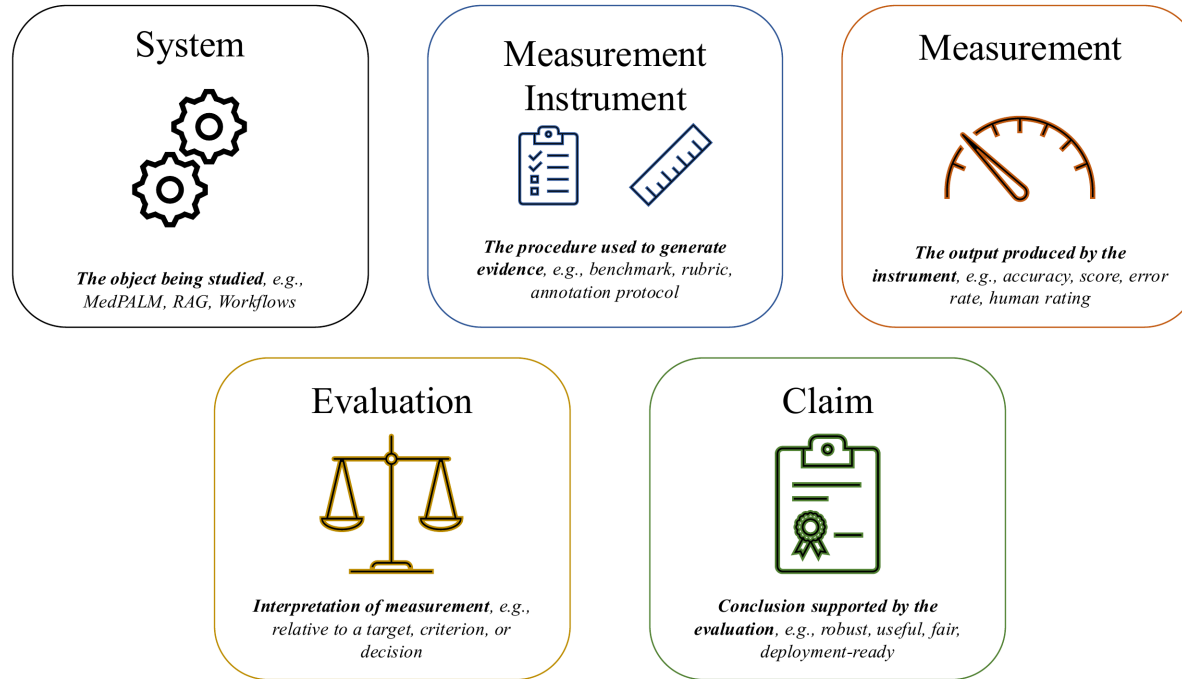
On Evaluating AI Methods vs. Models

What exactly are we measuring when we evaluate AI?

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Preliminary



The components of an evaluation: a system, a measurement instrument, the resulting measurement, its interpretation, and the supported claim.

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Motivation

Why the distinction matters

The basic problem

A benchmark is a measurement instrument (one of many).

But the same instrument can be used to evaluate completely different objects.

Sometimes we evaluate a fixed model.

Sometimes we evaluate a method for producing models.

These distinctions matter because the evidence needed to support a claim fundamentally changes depending on the object of evaluation.

A motivating example

Suppose we read:

Model A outperforms Model B on benchmark X.

What claim is actually being made?

- 1 A particular trained model is better on this benchmark?
- 2 Method A reliably produces better models than Method B?

Same result. Very different claims.

Thesis

Models and methods are different objects of evaluation [⁵].

A **model** is a concrete, realized artifact.

A **method** is a procedure for producing, adapting, or selecting artifacts.

So the same benchmark score can have entirely different meanings depending on what we are trying to measure.

[⁵]: Salaudeen, Olawale E., Florian E. Dorner, and Peter Hase. "On Evaluating Methods vs. Evaluating Models." *NeurIPS 2025 Workshop on LLM Evaluation* (2025).

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Models vs. Methods

Defining the Object of Evaluation

What is a method?

A method is a procedure or a mapping.

Examples

- ERM
- Domain adaptation
- LoRA finetuning
- Retrieval-augmented generation
- Clinical validation protocols

The object of evaluation is not one specific model. It is the reliable capability of a procedure to produce models --- the models themselves are part of the measurement instruments.

The Method Evaluation Regime

Suppose we want to compare method f and method g .

We are interested in their generalization from distribution P to distribution Q .

We apply both methods to data from P :

- $f(P, \cdot) \rightarrow \text{model } F_P$
- $g(P, \cdot) \rightarrow \text{model } G_P$

To evaluate if $f > g$, we test F_P and G_P on Q , where Q may or may not P .

Example. f : ResNet, g : AlexNet, P/Q : ImageNet.

Models as Measurement Instruments

In the method evaluation regime, what are we measuring?

The object of evaluation is **the methods f and g** .

Distributions P and Q , as well as the models F_P and G_P generated by the methods, are all merely **measurement instruments** --- *in many case, when you propose a new method, the method is the contributions not the models you built to show the method works.*

They are the lenses through which we observe the method.

What is a model?

A model is a concrete artifact.

Examples

- A fixed neural network checkpoint
- A sepsis prediction model deployed in one specific hospital
- A language model with a fixed prompt and decoding setup

The object of evaluation is tightly bounded and fixed.

The Model Evaluation Regime

Now suppose we are given two fixed models, F and G .

We want to evaluate how good they are, so we evaluate them on P .

In this regime:

- The models themselves are the objects of evaluation.
- They could have been trained on some P or something else entirely.
- Every dataset, choice, and random seed that went into making F and G matters --- variance in these components don't matter.

We are no longer evaluating a stable mapping under equal conditions. We are evaluating specific, realized systems.

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Generalization for Methods

The holdout method as a scientific instrument

The basic empirical pattern in machine learning development:

train on a dataset (from distribution P)



evaluate on held-out data (from distribution P)



compare by test performance

That is, you train and test on the **same distribution**, and you are using those results to **compare methods**.

Iterating on the same distribution

This same-distribution holdout set becomes an active engine for iteration:

evaluate on P → update method → train new model on P

We repeatedly compare methods on distribution P to find the best procedure.

But when evaluating methods, this process turns comparison into an empirical discipline about shift. It asks a fundamental question:

If a method iteratively developed and ranked on P beats its competitors...
Does that ranking comparison survive under Q ?

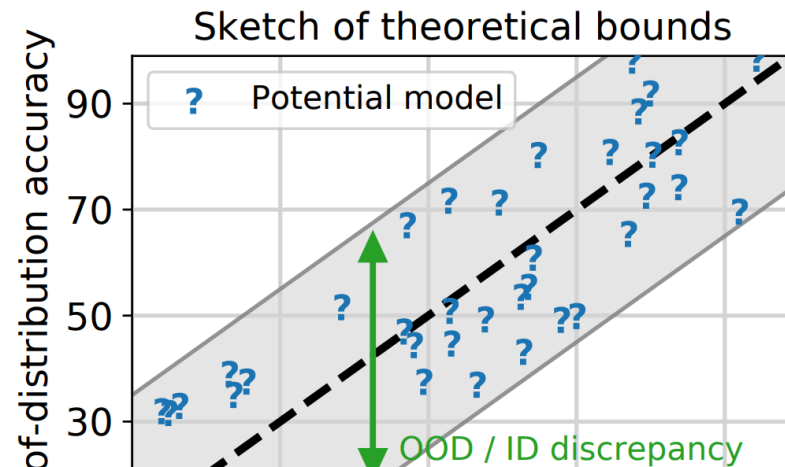
What is "Accuracy on the Line"?

Across many machine learning settings, a striking empirical phenomenon appears.

If you plot in-distribution (ID) accuracy against out-of-distribution (OOD) accuracy for a diverse family of models, they fall on a highly correlated linear trend.

higher ID accuracy $\approx$ higher OOD accuracy

When shifting to a new distribution, the absolute accuracy drops. But the relative improvements usually survive.



Accuracy on the line validates method development

This phenomenon is ultimately an observation about **method evaluation**.

It shows that a benchmark does not need to perfectly mirror a target clinical setting to be scientifically useful.

Because the linear trend holds, benchmarks successfully preserve **method rankings**.

It means we can confidently iterate on algorithms and develop new methods on standard benchmarks, knowing those relative improvements will likely transfer.

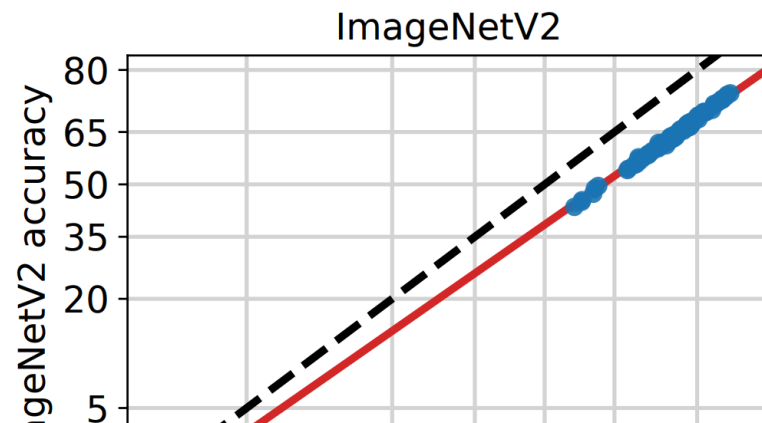
ImageNetV2: Fixed models on a resampled test set

ImageNetV2 asks what happens when we collect a new test set following the exact same data distribution process as the original ImageNet.

In this experiment, the objects being evaluated are **fixed, pre-trained models**.

- X-axis: A fixed model evaluated on the ImageNet test set.
- Y-axis: That *same* model evaluated on ImageNetV2.

Absolute accuracy drops, but the models fall on a line. The relative rankings of these artifacts are preserved under data resampling.

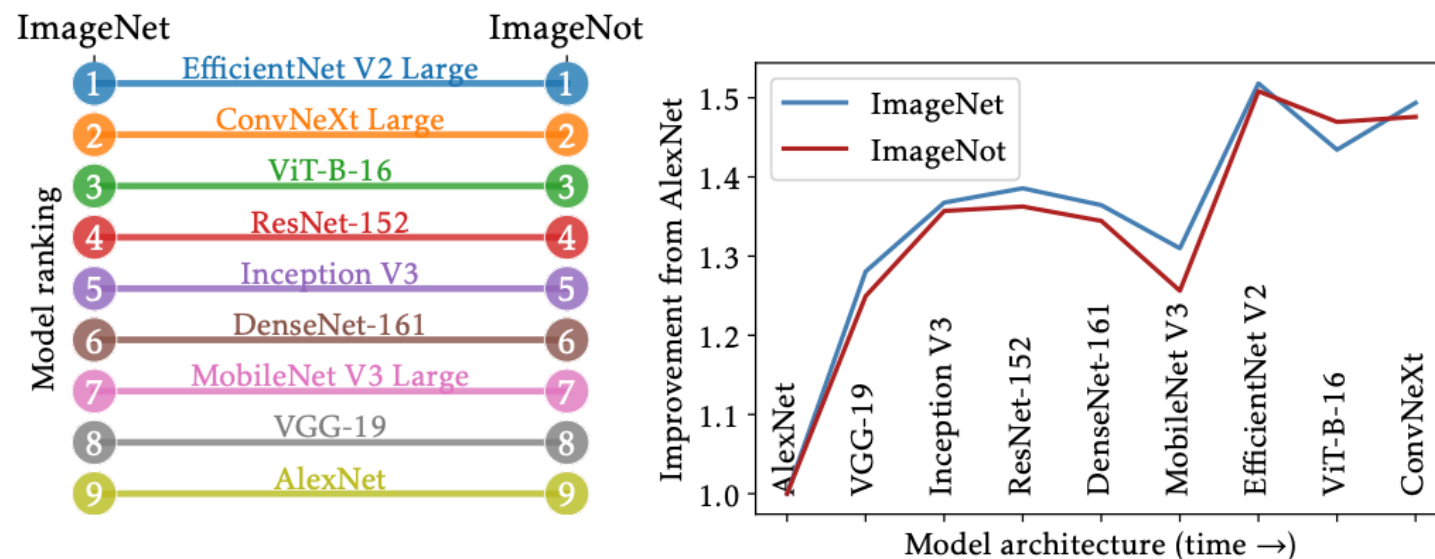


ImageNot: Retrained methods on a new distribution

ImageNot poses a much harder problem. It introduces a fundamentally different distribution with entirely different classes and images.

Crucially, we do not just transfer fixed ImageNet models. Instead, **methods are retrained from scratch** on ImageNot and evaluated on ImageNot.

This is a pure test of the method: does an architecture or training recipe that works well on ImageNet also reliably produce good models in a completely new environment?



What ImageNetV2 and ImageNot tell us together

ImageNetV2 shows that rankings survive simple test-set resampling.

But ImageNot shows something much more profound: **method rankings survive fundamental distribution shifts.**

Methods that improved over earlier architectures on ImageNet also improved when retrained on ImageNot.

The original benchmark captured something real and enduring about relative *method (model development) progress*, entirely independent of absolute performance.

The Method Takeaway

For method evaluation, accuracy-on-the-line is highly encouraging.

It suggests that benchmark progress reflects real method progress.

The claim isn't:

"The benchmark measures real-world accuracy."

The valid claim is:

"The benchmark preserves relative comparisons among methods under shift."

Can one push the distribution shift enough to where even this breaks? Maybe!

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Where the Line Breaks Down

From Methods back to Models

Why Accuracy-on-the-line is not enough for models

Accuracy-on-the-line evaluates the consistency of method rankings across a *family* of models.

It tells us:

Methods that rank higher on P tend to rank higher on Q .

But for a fixed model F , we need to know:

Does this specific artifact F that is better than G on P also better on Q ?

These are fundamentally different claims.

Accuracy on the Inverse Line

Performance ordering on distribution P can be **uncorrelated** or even **inversely correlated** with ordering on Q for fixed artifacts F_i :

- **Salaudeen et al. (2025)** characterize conditions under which one expects positive, negative, or no correlation and identify them in real-world problems. Under misspecification, shortcuts can actively invert between P and Q [¹].
- **Teney et al. (2022)** provide more empirical proof of inverse correlations on real-world datasets, often driven by models exploiting spurious features [²].
- **Sanyal et al. (2025)** also show that models can fall on the **"wrong line"**—where label noise and nuisance features mathematically shatter the expected positive correlation [³].

A clean method-level trend can hide catastrophic model-level failures.

[¹]: Salaudeen, Olawale, et al. "Are Domain Generalization Benchmarks with Accuracy on the Line Misspecified?" *Transactions on Machine Learning Research* (2025).

[²]: Teney, Damien, et al. "ID and OOD performance are sometimes inversely correlated on real-world

The breakdown of aggregate trends

Once F and G are fixed systems with their own idiosyncratic training histories, there is no guarantee the line will hold.

We can easily have a scenario where:

$$R_P(F) > R_P(G)$$

but due to specific artifact vulnerabilities:

$$R_Q(F) < R_Q(G)$$

Method rankings can be robust even when individual models are not.

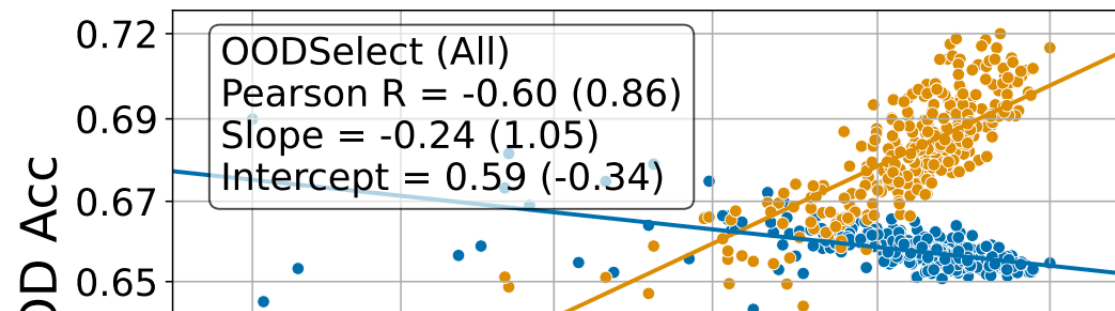
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Aggregation Hides OOD Failures

A central finding in **Salaudeen & Koyejo (2024)** is that the "Accuracy-on-the-Line" phenomenon is often a mathematical byproduct of aggregation that masks local failures.

Even when a method family shows a near-perfect correlation ($\rho \approx 1$) between ID and OOD performance in aggregate, that trend frequently vanishes or reverses when looking at **coherent subsets** of the shift.

- **The Observation:** Aggregate OOD trends are driven by the average shift across many environments.
- **The Finding:** Within a specific environment (a single hospital or demographic), the correlation can be zero or negative.



The Illusion of Progress: Simpson's Paradox

We find that "Accuracy-on-the-Line" can exhibit a form of **Simpson's Paradox** in Out-of-Distribution evaluation:

- 1 **At the Method Level:** Across a heterogeneous mix of many shifts, better ID models *appear* to be better OOD models on average.
- 2 **At the Model Level:** On a specific, "coherent" slice of that shift (e.g., a specific missingness pattern or clinical workflow), the relationship can be strictly **inverted**.

Key Finding: High aggregate correlation does not imply that ID improvements translate to OOD improvements for any specific, realized deployment setting.

Localized Shifts vs. Average Trends

A fixed clinical model F does not experience the "average" distribution shift; it experiences a **localized condition**.

Our analysis of various shifts shows:

- **Shortcut Reliance:** Aggregate trends are often buoyed by "easy" shifts, while the "hard" shifts—where shortcuts fail or invert—are buried in the mean.
- **The Reliability Gap:** An "optimal" model selected via the aggregate line may be the *worst* performing model for a specific hospital's demographic or scanner type.

Conclusion: To evaluate a model (the artifact), we must move from aggregate OOD metrics to **disaggregated OOD evaluation**.

[^4]: Salaudeen, Olawale, and Sanmi Koyejo. "Aggregation Hides Out-of-Distribution Failures." *Advances in Neural Information Processing Systems* (2024).

The Components of Model Validity

If a benchmark score is going to support a strong clinical or deployment claim, it must satisfy several layers of validity evidence.[^m2m]

- 1 Construct Validity
- 2 Content Validity
- 3 Criterion (Predictive) Validity
- 4 External Validity
- 5 Consequential Validity

[^m2m]: Salaudeen, Olawale, et al. "Measurement to Meaning: A Validity-Centered Framework for AI Evaluation." [arXiv:2505.10573](https://arxiv.org/abs/2505.10573) (2025).

1. Construct Validity

Are we actually measuring the latent trait we think we are measuring?

In AI, construct validity is frequently violated by **shortcut learning**.

If a model predicts pneumonia by relying on the specific type of X-ray scanner used rather than lung pathology, the benchmark score is measuring a spurious artifact, not the construct of "diagnostic capability."

How to view/assess:

- **Expert Assessment:** Engaging clinicians to review model explanations, feature importance, and failure modes to verify alignment with established pathophysiological reasoning rather than spurious correlations.
- **Counterfactual testing:** Altering background artifacts (like removing a surgical marker) to see if the prediction flips.
- **Saliency maps & feature attribution:** Visualizing what pixel regions drive the model's decision.
- **Concept bottleneck models:** Forcing the model to predict interpretable clinical concepts before

2. Content Validity

Does the evaluation adequately cover the necessary domain?

A clinical claim usually implies broad usefulness. If an evaluation dataset only contains data from young, healthy adults, or lacks severe disease presentations, it has low content validity.

How to view/assess:

- **Epidemiological mapping:** Comparing the distribution of the test set against the real-world prevalence of the disease.
- **Disaggregated metrics:** Slicing performance across intersectional demographics, comorbidities, and severity levels.
- **Expert domain review:** Having clinicians audit the inclusion and exclusion criteria of the benchmark dataset.

3. Criterion / Predictive Validity

Does the benchmark performance actually predict downstream utility?

A model might have a high AUROC on a retrospective dataset, but does that score correlate with better clinical decisions in the workflow?

Predictive validity requires evidence that the measured score translates into the intended real-world impact (e.g., reduced time-to-treatment or improved patient outcomes).

How to view/assess:

- **Shadow mode validation:** Running the model silently on live hospital data to compare predictions against actual clinician decisions.
- **Clinician-in-the-loop A/B testing:** Measuring whether doctors assisted by the AI make better decisions than those without it.
- **Decision curve analysis:** Evaluating the net clinical benefit of the model across different risk

4. External Validity

Does the entire validity argument hold across different settings?

If a model is proven safe and effective in Hospital A, can we transfer that claim to Hospital B?

External validity requires understanding the scope conditions of the model. What demographic, temporal, or workflow shifts cause the preceding validity arguments to break down?

How to view/assess:

- **Multi-center validation:** Testing the model on distinct geographic populations and different hospital systems.
- **Temporal validation:** Training on historical data (e.g., 2018-2022) and explicitly testing on future data (e.g., 2024-2026) to check for practice drift.
- **Hardware shift testing:** Evaluating performance across different acquisition devices (e.g., Siemens vs. GE scanners).

5. Consequential Validity

What are the systemic effects and potential harms of deploying the model?

Deployment changes the environment. Even a model with high predictive validity can fail here.

Does the model cause clinician alert fatigue? Does it inadvertently allocate resources away from vulnerable populations? Consequential validity demands evidence that the overall social and systemic impact of using the model justifies its deployment.

How to view/assess:

- **Workflow auditing:** Surveying clinicians to measure cognitive load, automation bias, and alert fatigue post-deployment.
- **Resource allocation tracking:** Monitoring if the algorithm systematically shifts care access or wait times away from marginalized groups.
- **Post-market surveillance:** Continuous tracking of longitudinal patient outcomes to catch delayed

Validity as a Burden of Proof

A model-level claim:

This model predicts sepsis with high AUROC on similar distributions. *(Requires external and consequential validity evidence)*

A use claim:

This model improves well-defined clinical decisions. *(Requires predictive, external, and validity)*

A deployment claim:

This model safely improves patient outcomes across the health system. *(Requires external, construct, content, and consequential validity)*

The Proper Role of Benchmarks

If the validity burden for clinical use is so high, what are benchmarks actually good for?

Benchmarks face an incredibly high bar to "mean" something for clinical deployment. However, they are essential for two upstream tasks:

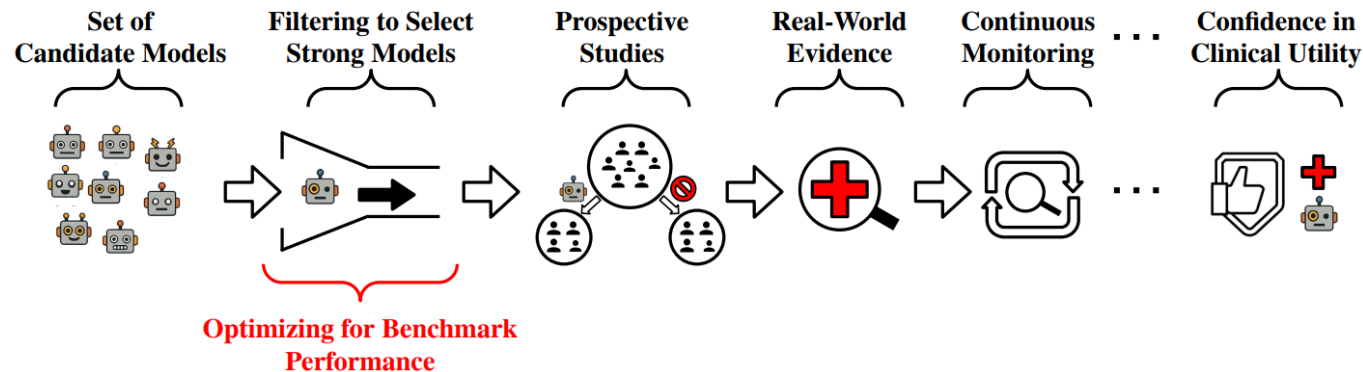
- 1 **Hill-Climbing:** Iterative method development. Benchmarks are the engine for optimizing procedures and comparing methods (the $f > g$ regime).
- 2 **Triaging:** Filtering out bad artifacts. If a model fails on a well-constructed benchmark, it is not worth the cost and risk of clinical validation.

You Cannot Benchmark Your Way to Clinical Use

Benchmarks abstract away the environment to create standardized measurement. But clinical deployment requires evaluating the artifact *inside* the environment.

A benchmark can tell you a model has basic predictive capability (Criterion Validity).

But a benchmark **cannot** validate workflow fit, clinician reliance, or patient outcomes. Those require exiting the benchmark paradigm entirely.



Benchmark optimization is only the first filter; clinical utility requires prospective studies, real-world evidence, and continuous monitoring. [⁴]

Beyond the Benchmark: Evaluating Models

To validly evaluate models for clinical use, we need the "other things" outside the holdout set:

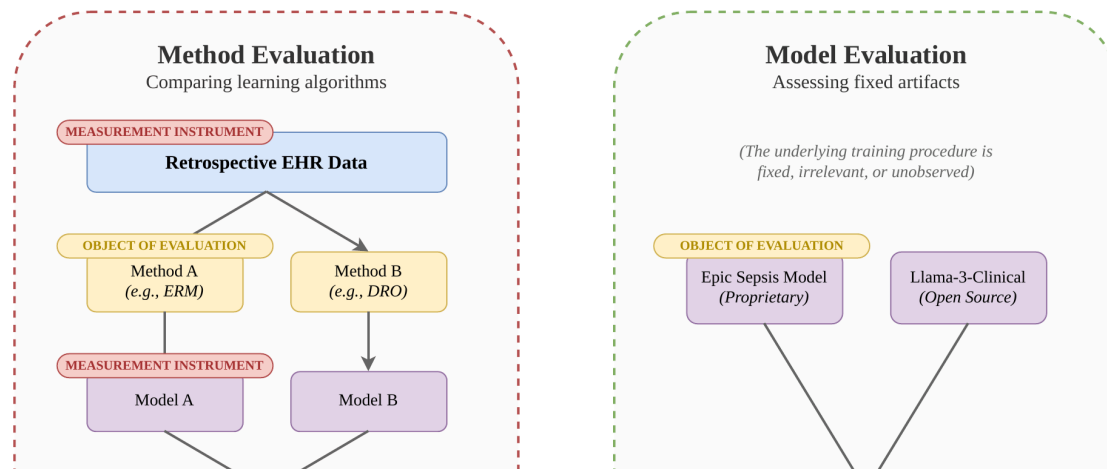
- **Local Retrospective Validation:** Testing the artifact on the specific hospital system's historical data to establish a local baseline.
- **Shadow Mode / Prospective Evaluation:** Running the model silently on live data to test predictive validity against actual, real-time clinician decisions.
- **Human-AI Interaction Trials:** Measuring how the artifact actually alters clinician behavior (Consequential Validity).
- **Continuous Monitoring Infrastructures:** Tracking performance post-deployment to catch the inevitable distribution shifts.

You build methods on benchmarks. You validate models in the real world.

Conclusion: One Benchmark, Two Paths

The scientific value of a benchmark score depends entirely on the path of interpretation:

- **The Method Path (Upstream Discovery):** Benchmarks are highly valid for hill-climbing, comparing algorithms, and tracking progress in model development. Accuracy-on-the-line suggests these rankings are robust.
- **The Model Path (Downstream Deployment):** Benchmarks are insufficient for clinical claims. They serve only as a "triage" filter. Validity for clinical utility must be built outside the benchmark, through local, disaggregated, and prospective evaluation of the artifact.



Takeaways

- 1 **Discovery vs. Verification:** Benchmarks are for finding the best *procedure* (discovery); clinical evaluation is for verifying the specific *artifact* (verification).
- 2 **Robustness is Not Safety:** Aggregate method rankings are often robust under shift, but individual artifacts are idiosyncratic and can actively fall on the inverse line.
- 3 **The Limits of the Holdout Set:** You cannot benchmark your way to clinical utility. Task performance is a narrow criterion; clinical usefulness is a broad construct that includes workflow and human interaction.
- 4 **The Validity Burden:** As a claim moves from "it performs well on a test set" to "it should be used on patients," the burden of proof shifts from scores to a multi-faceted validity argument.

Closing: Moving Beyond the Score

Before interpreting any AI result, we must shift the conversation from "What is the accuracy?" to "What is the object and the claim?"

Are we evaluating a **method** (for progress) or an **model** (for use)?

Is the benchmark used for **hill-climbing** or is it being over-interpreted as **clinical proof**?

Does the evidence support a narrow **criterion** or a broad **clinical construct**?

We build methods on benchmarks, but we must validate models in the real world.

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Thank you

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Questions?